

Service Menu Optimization for Many-to-One Demand-Responsive Transit with Meeting Points and Pickup Time Windows

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Abstract

We study service-menu optimization for many-to-one demand-responsive transit (DRT), where each passenger-facing offer is a bundle of a meeting point and a pickup time window. The display decision is formulated as a Multinomial Logit (MNL) assortment problem with operationally endogenous bundle costs: predicted routing cost, capacity risk, pickup timing, and price jointly determine menu value. The solver uses exact subset enumeration for small candidate sets and greedy forward selection for larger sets, with exact-vs-greedy diagnostics quantifying approximation quality. The empirical program is tiered: the RC (Clustered-Random) benchmark supports exact-gap and robust-filtering mechanism diagnostics, low/medium/high uptake regimes identify when menu design is behaviorally active, and Austin/Seattle provide external directional checks. Filtering based on estimated time of arrival (ETA) can reshape the feasible meeting-point assortment before optimization acts; exact-small/greedy-large menu construction yields a transparent operational approximation whose gap is empirically auditable; and pricing/menu effects are meaningful only in regimes where acceptance and non-home uptake are observable rather than nearly degenerate.

Keywords: demand-responsive transit, service menu optimization, assortment optimization, multinomial logit, robust filtering, pickup time windows

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1. Introduction

Many-to-one demand-responsive transit (DRT) systems must translate operational flexibility into passenger-facing offers that are both understandable and executable. In this paper, the offer shown to a passenger is not a single pickup promise but a *service menu*: a set of candidate bundles, each specifying a meeting point and a pickup time window [17, 10]. The operator must decide which bundles to display before observing the passenger’s choice, and the selected bundle then enters the routing layer.

This display decision is an assortment problem with a transportation twist. Classical assortment models treat products as exogenous items with known utilities and revenues. In DRT, by contrast, a service bundle has operationally endogenous attributes: its attractiveness depends on walking distance, pickup timing, in-vehicle time, and price, while its cost depends on current vehicle routes, capacity, and downstream re-optimization [1]. Showing all feasible bundles can dilute demand steering and expose noisy offers; showing too few can collapse the menu to home pickup and remove profitable meeting-point alternatives before the passenger ever sees them.

The central research question is therefore not simply which heuristic wins in one calibration. We ask how a DRT platform can formulate, solve, and diagnose the service-menu assortment problem when bundle values are predicted online and candidate eligibility depends on uncertain ETA signals. This question is managerial as well as algorithmic: if a filtering rule removes most feasible meeting-point bundles before pricing or menu optimization acts, the downstream algorithm may look weak even when the real failure is upstream candidate-set distortion.

This paper makes four contributions. First, we formulate the many-to-one DRT display decision as a service-bundle assortment optimization problem under MNL choice, with meeting-point and pickup-window attributes, an outside option, and route-aware surrogate costs. Second, we implement an exact-

small/greedy-large solution framework: exact subset enumeration provides a benchmark on small candidate sets, while greedy forward selection provides a scalable approximation for larger sets (mechanism-diagnostic evidence). The pipeline logs exact-vs-greedy value gaps, menu overlap, and build times so approximation quality is empirically auditable rather than asserted. Third, we treat ETA filtering as a diagnostic eligibility comparison rather than a fixed pre-processing detail, comparing hard, calibrated, interval-overlap, and no-filter variants under a paired evaluation contract (mechanism-diagnostic evidence). Fourth, we organize the evidence by behavioral regime: low-, medium-, and higher-uptake RC calibrations identify when menu design is behaviorally live, while Austin and Seattle remain external directional checks rather than broad validation claims (behavioral stress-test and descriptive evidence).

The resulting contribution is deliberately narrower than a universal dominance claim. The paper develops a reproducible optimization-and-diagnostics framework for DRT service menus, shows how exact benchmarks and robust filtering diagnostics can strengthen simulation evidence, and clarifies when profit, uptake, and welfare comparisons are meaningful.

2. Related Work

2.1. Demand-Responsive Transit and Dial-a-Ride

Demand-responsive transit (DRT) systems have been studied extensively under the dial-a-ride problem (DARP) framework, in which passengers submit pickup and delivery requests and a fleet of vehicles must be routed to serve them subject to time-window and capacity constraints. Cordeau and Laporte [5] provide a comprehensive survey of DARP models and exact algorithms. Parragh et al. [15] extend this to the broader class of pickup-and-delivery problems. Ho et al. [12] survey more recent developments, including dynamic and real-time variants relevant to operational DRT deployments.

Dong et al. [9] extend the DARP to include passenger accept/reject decisions based on service utilities, a formulation closely related to our MNL-based choice

model; their work motivates the inclusion of an explicit outside option in the choice model (Section 4.2).

The many-to-one variant studied here — where all passengers share a common destination — arises naturally in first-mile/last-mile transit and commuter shuttle contexts. Meeting-point strategies, in which passengers walk to a shared pickup location rather than being collected door-to-door, have been shown to substantially improve vehicle efficiency [17]. High-capacity ride-sharing with dynamic trip-vehicle assignment further demonstrates that pooling passenger requests at shared stops can achieve near-optimal fleet utilization [2], while minimum-fleet analysis confirms that smart assignment of shared rides substantially reduces the number of vehicles required to serve a given demand [20]. Our work extends this line by treating the *set* of displayed meeting-point offers as a decision variable rather than a fixed system parameter.

Recent work has extended meeting-point strategies in several directions. Czoska et al. [7] develop methods for identifying real-world meeting points in shared DRT systems using geographic and accessibility criteria. Cortenbach et al. [6] formulate the dial-a-ride problem with meeting points as a mixed-integer program, providing exact solutions for small instances and heuristics for larger ones. Stumpe et al. [18] study the design of taxi ridesharing systems with shared pick-up and drop-off locations, finding that meeting-point flexibility substantially reduces vehicle-kilometres travelled. Zheng et al. [25] analyze meeting points in flex-route transit services and show that introducing passenger walking can improve service productivity under appropriate operating conditions. Our work complements these contributions by focusing on the *display* decision: given a set of feasible meeting-point bundles, which subset should be shown to the passenger?

2.2. Assortment Optimization under MNL

The problem of selecting which products to display to a customer who chooses according to an MNL model is known as assortment optimization. Talluri and van Ryzin [19] establish that revenue-ordered assortments are optimal

under the uncapacitated MNL model and derive the Lambert-W pricing formula that we adopt in Section 4.2. Rusmevichientong et al. [16] extend this to capacity-constrained and dynamic settings, showing that greedy revenue-ordered policies remain near-optimal under mild conditions.

Wang [22] extend the assortment problem to the capacitated setting, showing that the revenue-ordered property breaks down when products have heterogeneous costs — a result directly relevant to our setting where bundles have heterogeneous operational costs. Désir et al. [8] establish hardness and approximation results for capacitated assortment optimization, further illustrating why constrained assortment problems generally require more than simple ordering rules.

Our setting differs from classical assortment optimization in two important respects. First, the “products” (service bundles) are generated dynamically per request and their attributes are predicted rather than known. Second, the operator’s objective couples menu design with vehicle routing costs, so the revenue-maximising assortment and the profit-maximising menu can diverge. The menu-value surrogate in Eq. (6) is designed to capture this coupling through the system evaluation cost $\hat{c}_{\text{sys}}(b)$.

2.3. Ride-Sharing and On-Demand Mobility

Agatz et al. [1] survey optimization approaches for dynamic ride-sharing, highlighting the tension between real-time assignment and service quality. Zha et al. [23] analyze how pricing mechanisms affect driver supply and passenger demand in ride-sourcing, showing that platform-level price adjustments have heterogeneous effects on service quality and operator revenue. Fielbaum et al. [10] study on-demand ridesharing with optimized pickup and drop-off walking locations, a setting closely related to ours; their results confirm that allowing flexible pickup locations significantly expands the feasible service region and improves system efficiency.

Our contribution is complementary: rather than optimizing the *location* of meeting points, we study which subset of pre-generated meeting-point offers to

display to the passenger, and how the display decision interacts with MNL-based pricing and passenger choice.

2.4. Vehicle Routing and Route Optimization

The back-end route optimizer in our system uses the Hybrid Genetic Search (HGS) algorithm of Vidal [21], which provides high-quality solutions to capacitated vehicle routing problems with time windows. We treat the routing layer as a black box and focus on the menu-policy layer; the separation between these layers is a deliberate design choice that allows menu policies to be evaluated without re-solving the full VRPTW at every decision epoch.

2.5. Positioning of This Work

The present paper sits at the intersection of assortment optimization and DRT operations. Relative to prior work, it brings together the menu-display decision in a many-to-one DRT system with a three-layer framework combining MNL-based passenger choice, common-offset pricing, route-aware menu-value evaluation, and exact-small/greedy-large assortment solving. The paired methodology separates approximation diagnostics, robust filtering behavior, uptake-regime sensitivity, and cross-instance directional checks, making the filtering layer and the greedy approximation empirically auditable rather than hidden implementation details. The tension between platform profit, pooling decisions, pricing, and passenger welfare that motivates our uptake-regime analysis has been noted in ride-sharing equilibrium models [24], and stated-preference and hybrid-choice approaches to demand-adaptive and on-demand transit service design [11, 3] underscore the importance of modeling passenger choice explicitly when evaluating menu policies.

3. Problem Setting

3.1. System Overview

We consider a many-to-one demand-responsive transit (DRT) system in which passengers are collected from dispersed origins and transported to a com-

mon destination (e.g., a transit hub or workplace zone). At each decision epoch, a passenger submits a pickup request. The operator must decide which service offers to display, after which the passenger selects one displayed offer or opts out. The selected offer is then executed by the vehicle routing layer.

The distinguishing feature of this setting relative to standard dial-a-ride formulations [5] is that passengers are not simply assigned to vehicles; instead, they choose among a *displayed menu* of service bundles. The operator therefore faces a joint problem of offer design and demand steering.

3.2. Notation and Entities

Let $\mathcal{R}$ denote the set of passenger requests arriving over a planning horizon T . For each request $r \in \mathcal{R}$, the simulator first constructs a candidate set of feasible bundles, denoted $\mathcal{B}(r)$. We use *feasible bundle set* as the primary term and *candidate set* as a shorthand for the same object. A *service bundle* $b \in \mathcal{B}(r)$ is a tuple

$$b = (m_b, [t_b^{\text{lo}}, t_b^{\text{hi}}], p_b),$$

where m_b is a meeting point (a geographic location the passenger walks to for pickup), $[t_b^{\text{lo}}, t_b^{\text{hi}}]$ is the offered pickup time window, and $p_b \in \mathbb{R}$ is the price charged to the passenger. A special *home bundle* b^0 always satisfies $m_{b^0} = r.\text{origin}$ and imposes zero walking distance. A *displayed menu* is a subset $\mathcal{M}(r) \subseteq \mathcal{B}(r)$. For limited-menu policies, k denotes the cap on displayed non-home offers, so $|\mathcal{M}(r) \setminus \{b^0\}| \leq k$. Passengers may also choose an *outside option*, meaning that no bundle in $\mathcal{M}(r)$ is selected.

Table 1 summarizes the main symbols used throughout the paper. The formal problem statement is given in Section 3.3.

Note on notation. To avoid collision with the scoring weights $\mu_m, \mu_w, \mu_t, \mu_v$ introduced in Section 4.3, we use β (not λ) for the passenger price sensitivity throughout. The symbol λ is reserved for the Lambert W pricing derivation in Section 4.2, where it denotes the same quantity as β but in the context of the pricing formula.

Predicted versus realized quantities. Symbols with hats denote predictor outputs or menu-construction surrogates, such as $\hat{c}(b)$, $\hat{\tau}_b$, $\hat{l}_b$, and $\hat{c}_{\text{sys}}(b)$. Unhatted outcome quantities refer to realized evaluation outcomes under replayed request traces unless a table or caption explicitly labels them as predictor diagnostics.

False-negative pruning. For a filtered menu policy, we define the false-negative pruning rate on request r as the share of feasible non-home bundles in $\mathcal{B}(r) \setminus \{b^0\}$ that are removed by the ETA pre-filter:

$$\text{FN}(r) = \frac{|\mathcal{P}_{\text{FN}}(r)|}{|\mathcal{B}(r) \setminus \{b^0\}|},$$

where $\mathcal{P}_{\text{FN}}(r) \subseteq \mathcal{B}(r) \setminus \{b^0\}$ denotes the feasible non-home bundles pruned before menu scoring. All reported false-negative pruning rates use this non-home feasible-bundle denominator.

3.3. Formal Problem Statement

Problem Statement (Menu Design for DRT).

Given:

- A set of passenger requests $\mathcal{R}$ arriving over planning horizon T .
- For each request $r \in \mathcal{R}$, a feasible bundle set

$$\mathcal{B}(r) = \{b = (m_b, [t_b^{\text{lo}}, t_b^{\text{hi}}], p_b) \mid \text{feasibility}(b, r)\},$$

where feasibility requires: (i) the pickup time window $[t_b^{\text{lo}}, t_b^{\text{hi}}]$ satisfies the passenger’s latest acceptable arrival, (ii) the assigned vehicle has sufficient remaining capacity, and (iii) the meeting point m_b is reachable within the vehicle routing time windows enforced by the back-end optimizer.

- A cap $k \geq 1$ on displayed non-home offers.
- An MNL choice model with parameters $(\bar{u}, u^{\text{home}}, \alpha_\tau, \alpha_\Delta, \sigma, \beta)$ (Table 2) and an implemented common-offset pricing parameter δ_{impl} (Section 4.2).

Decision: For each request r , select a displayed menu

$$\mathcal{M}(r) \subseteq \mathcal{B}(r), \quad |\mathcal{M}(r) \setminus \{b^0\}| \leq k.$$

Objective: Maximize aggregate expected net profit over all requests:

$$\max_{\{\mathcal{M}(r)\}_{r \in \mathcal{R}}} \Pi = \sum_{r \in \mathcal{R}} \sum_{b \in \mathcal{M}(r)} \pi(b \mid \mathcal{M}(r)) (p_b - c(b)),$$

where $\pi(b \mid \mathcal{M}(r))$ is the MNL choice probability (Eq. 2) and $c(b)$ is the marginal operational cost of serving bundle b .

Hardness note: The exact solution requires evaluating all $\binom{|\mathcal{B}(r)|}{k}$ subsets per request, each requiring a joint pricing and menu-value evaluation. This is practical for small candidate sets and useful as a benchmark, but combinatorially expensive at scale. We therefore use exact enumeration as a small-set reference and greedy forward selection as the scalable approximation described in Section 4.3.

3.4. Three-Layer Decision Structure

The benchmark environment decomposes each request episode into three sequential layers.

Layer 1 — Candidate generation. The simulator identifies all feasible meeting-point neighborhoods for request r and constructs the bundle set $\mathcal{B}(r)$, including the home bundle b^0 . Feasibility is assessed using approximate front-end pickup windows; exact vehicle routing time windows (VRPTW) are enforced by the back-end route optimizer [21] after a bundle is selected.

Layer 2 — Menu policy. The operator selects a menu $\mathcal{M}(r) \subseteq \mathcal{B}(r)$ to display. This is the decision layer studied in this paper. All competing policies in our experiments share the same Layer 1 candidate generator and Layer 3 choice model; they differ only in how they construct $\mathcal{M}(r)$.

Layer 3 — Passenger choice. The passenger selects one bundle from $\mathcal{M}(r)$ according to a Multinomial Logit (MNL) model [4, 13]. The selected bundle is routed by the vehicle layer.

3.5. Objective

The operator’s objective is to maximize aggregate net profit over all requests:

$$\Pi = \sum_{r \in \mathcal{R}} \mathbb{E}[p_{\tilde{b}} - c(\tilde{b})],$$

where $\tilde{b}$ is the bundle chosen by the passenger from $\mathcal{M}(r)$, $p_{\tilde{b}}$ is the price charged, and $c(\tilde{b})$ is the marginal operational cost. In practice, $c(b)$ captures travel cost, discount cost, service cost, and home-pickup failure cost. The menu policy influences Π both by restricting which bundles can be chosen and by shaping the prices set for each displayed bundle.

4. Method

4.1. Shared Bundle Predictor

A shared predictor is trained once per data split using the *exhaustive full-display* regime, in which all feasible bundles $\mathcal{B}(r)$ are shown to every passenger. The core menu objects $\mathcal{B}(r)$, $\mathcal{M}(r)$, b^0 , and the outside option are defined in Section 3.2. The predictor estimates three bundle-level operational signals from request and network features: predicted marginal cost $\hat{c}(b)$, predicted pickup ETA $\hat{\tau}_b$, and predicted in-vehicle time $\hat{t}_b$. Frozen evaluation then reuses this checkpoint while varying only the menu policy, isolating the effect of menu construction from any confound introduced by training a different predictor per policy.

Architecture.. The predictor is a two-layer convolutional neural network (CNN-2d) that encodes the spatial state of the service area as a $2 \times 11 \times 11$ grid tensor (2 temporal layers, 11×11 spatial grid). The convolutional backbone consists of two Conv2d layers (16 and 32 filters, 3×3 kernels, same padding) followed by average pooling, producing a flattened spatial embedding. A 4-dimensional auxiliary feature vector (remaining vehicle capacity and related scalars) is concatenated before three fully connected layers ($256 \rightarrow 128 \rightarrow 3$ outputs). The three output heads predict $\hat{c}(b)$, $\hat{\tau}_b$, and $\hat{t}_b$ jointly. Training uses

the Huber loss ($\delta = 1.0$) with the Adam optimizer (learning rate 10^{-3} , replay buffer of 500 episodes).

Input features.. The spatial grid encodes the current demand density and vehicle occupancy at each grid cell across two time intervals. The auxiliary vector encodes the remaining capacity of the candidate vehicle, the time elapsed in the current episode, and two bundle-specific scalars (walking distance and candidate meeting-point index). The predictor is trained on (s_t, b, y_t) tuples collected under the full-display policy, where $y_t = (c_t, \tau_t, \iota_t)$ are the realized cost, pickup time, and in-vehicle time for the selected bundle.

Prediction accuracy.. Predictor MAE on a held-out 20% partition of the training buffer is reported in Appendix Appendix D. The raw CNN cost head improves on a naive mean baseline, while ETA and IVT heads are blended at inference time with heuristic routing signals (blend weight 0.35 CNN, 0.65 heuristic). Because menu filtering acts on the blended signal rather than the raw CNN head, the paper distinguishes raw-head error from the effective signal diagnostics reported in Section 6.

Deployed, predicted, and realized time quantities.. The implementation uses three distinct time quantities that should not be conflated. The *predicted* quantities ($\hat{\tau}_b, \hat{\iota}_b$) are the model-based bundle attributes passed into utility scoring and menu construction. The *deployed filtering signal* is the ETA quantity actually thresholded by the implementation when ETA pre-screening is enabled; in the deployed policy this is the blended ETA signal described above, while companion variants replace that signal with heuristic, stronger-calibrated, or oracle-style proxies. The *realized* quantities are the simulator outcomes observed after a passenger choice is made and the route is executed. Reported net profit, acceptance, walking, pickup deviation, and in-vehicle time outcomes are therefore realized performance metrics, whereas the ETA/IVT MAE diagnostics in Section 6 evaluate the signal that was displayed or filtered.

4.2. MNL Choice Model and Implemented Pricing Rules

Passenger choice follows a Multinomial Logit model [4, 13]. Given a displayed menu $\mathcal{M}(r)$, the utility of bundle b for passenger r is

$$u_b = \bar{u} + \mathbb{1}[b = b^0] \cdot u^{\text{home}} + \alpha_\tau \hat{t}_b + \alpha_\Delta \Delta_b - (1 - e^{-d_b/\sigma}) + \beta p_b, \quad (1)$$

where $\bar{u}$ is a base utility intercept, $u^{\text{home}} > 0$ is the additional utility of the home bundle, $\hat{t}_b$ is predicted in-vehicle time, $\Delta_b = \hat{\tau}_b - t_r^{\text{req}}$ is the pickup time deviation, d_b is walking distance, $\sigma > 0$ is a distance-scale parameter measured in the same distance units as the underlying instance, and $\beta < 0$ is the passenger price sensitivity. The saturating penalty $(1 - e^{-d_b/\sigma})$ equals zero at $d_b = 0$ and approaches -1 as $d_b \rightarrow \infty$, ensuring that walking disutility is monotone and bounded.

Table 2 reports the numerical values of all MNL utility parameters used in the experiments. These values are fixed design parameters of the simulator, not estimates from passenger survey data; they encode one stylized commuter-DRT preference regime and are held constant within each reported study.

The choice probability for bundle $b \in \mathcal{M}(r)$ is

$$\pi(b \mid \mathcal{M}) = \frac{e^{u_b}}{e^{u_0} + \sum_{b' \in \mathcal{M}} e^{u_{b'}}}, \quad (2)$$

where $u_0 = 0$ is the normalized utility of the outside option (no bundle chosen). Passengers may therefore reject the displayed menu entirely.

Parameter plausibility ranges. The MNL parameters in Table 2 are fixed simulator design values rather than estimates from passenger surveys or revealed-preference data. However, two key parameters fall within plausibility ranges reported in the DRT and stated-preference literature. The price sensitivity $\beta = -0.25$ is consistent with demand-responsive and on-demand transit studies that estimate fare sensitivity as part of stated-preference or hybrid-choice models [11, 3]. The in-vehicle time weight $\alpha_\tau = -0.002$ corresponds to a low but non-trivial time valuation in the stylized utility units, compatible with the range of IVT coefficients in Ben-Akiva and Lerman [4] and related discrete-choice analyses. The outside-option utility $u_0 = 0$ (normalized) is tested across a symmetric

scan in Section 6.4; this scan spans $u_0 \in \{-1.0, -0.5, 0.0, +0.5, +1.0\}$ to assess how the menu-optimization results respond to the attractiveness of opting out. Walking-distance sensitivity and home-bundle bonus lack direct literature anchors and remain uncalibrated design parameters; their sensitivity is partially captured by the regime-level MNL stress tests reported in Section 6.4. This framing constitutes literature-bounded sensitivity testing, not external demand validation with revealed preferences or survey data.

Pricing transforms evaluated. The pricing-baseline study keeps the same menu-construction logic but swaps the displayed-price transform. Three pricing rules are evaluated; they share the same predicted costs, feasible-set logic, route-aware scoring, and clipping constraints, so that the comparison isolates how much the pricing transform itself contributes once filtering and menu composition are held fixed. The price-sensitivity parameter β governs how strongly passengers react to displayed prices, and the platform-level tradeoff between revenue extraction and demand stimulation is a recurring theme in ride-sourcing pricing analysis [23].

Cost-plus pricing. The simplest rule removes the shared-offset optimization entirely:

$$p_b = \text{clip}(\hat{c}(b) - R, p_{\min}, p_{\max}),$$

where $\hat{c}(b)$ is the predicted marginal cost and R is a revenue target. Each displayed price equals predicted cost minus the revenue target, clipped to the platform bounds $[p_{\min}, p_{\max}]$. This is the most transparent pricing rule: no demand-side parameter enters the price calculation. It serves as a baseline showing what happens when demand-aware price adjustment is removed, isolating the contribution of menu construction and filtering alone.

Flat-markdown pricing. A single constant offset is applied to every displayed bundle:

$$p_b = \text{clip}(\hat{c}(b) + \kappa, p_{\min}, p_{\max}),$$

where κ is one common constant calibrated per regime. Like cost-plus, the flat-markdown rule uses no demand model; unlike cost-plus, it applies a regime-

tuned markdown rather than a cost-minus-target formula. Notably, the flat-markdown rule outperforms the Lambert-W transform in some calibrated uptake regimes, suggesting that the closed-form optimization does not always dominate simpler alternatives and that the practical value of the Lambert-W offset is regime-dependent.

Lambert-W common-offset pricing. The third rule prices a displayed menu through a shared offset derived from the Lambert-W function. For each bundle, $p_b^{\text{ref}}(\delta) = \hat{c}(b) - R - \delta$. Let u_b^0 denote the utility of b evaluated at zero displayed price. The code constructs

$$S_{\text{impl}} = e^{u_0} + \sum_{b \in \mathcal{M}} e^{u_b^0 + \beta(\hat{c}(b) - R)},$$

and solves

$$\delta_{\text{impl}} = \frac{W_0(S_{\text{impl}}/e) + 1}{\beta}, \quad (3)$$

followed by clipped displayed prices

$$p_b = \text{clip}(\hat{c}(b) - R - \delta_{\text{impl}}, p_{\min}, p_{\max}). \quad (4)$$

This is an implemented common-offset pricing heuristic, not the exact optimizer of the simulator’s clipped heterogeneous-cost profit objective. The full derivation is provided in Appendix Appendix B. The home bundle is retained in the pricing set (`menu_keep_home=True`) for numerical stability because removing it sharply reduces the exponential mass in S_{impl} and tends to collapse all OOH prices to the floor.

4.3. Greedy Menu Construction and System Evaluation

System evaluation cost.. Menu construction uses a route-aware surrogate cost for each displayed bundle,

$$\hat{c}_{\text{sys}}(b) = \hat{c}(b) + \underbrace{\mu_r \cdot \kappa \cdot \rho_b}_{\text{route-delay penalty}} + \underbrace{\frac{\mu_q}{q_b + \epsilon}}_{\text{capacity-risk penalty}}, \quad (5)$$

where ρ_b is the predicted route-delay increment, q_b is remaining vehicle capacity, κ is a cost multiplier, and $\epsilon > 0$ is a numerical-stability constant.

Menu-value surrogate.. After applying the common-offset pricing heuristic to a candidate menu, the implementation evaluates that menu with the surrogate

$$\hat{\Pi}(\mathcal{M}) = \sum_{b \in \mathcal{M}} \pi(b \mid \mathcal{M}) (p_b - \hat{c}_{\text{sys}}(b)), \quad (6)$$

where $\pi(b \mid \mathcal{M})$ is the MNL probability from Eq. 2 and p_b is the clipped displayed price from Eq. 4. This surrogate is not the exact global DRT objective, but it is the criterion actually used by the implemented menu-selection code.

Exact-small/greedy-large menu solver.. In the main `menu_optimization` policy, the home bundle is retained and the operator builds a menu of up to k non-home offers. The solver is controlled by `menu_selection_solver`. In **exact** mode, candidate sets no larger than `menu_exact_threshold` are solved by enumerating all subsets and selecting the menu maximizing Eq. 6. In **greedy** mode, the policy always uses forward selection. In **auto** mode, the implementation uses exact enumeration for small feasible sets and greedy forward selection otherwise:

1. start from the home bundle;
2. select the single non-home offer b that maximizes $\hat{\Pi}(\{b^0, b\})$;
3. iteratively add the remaining offer that yields the largest increase in $\hat{\Pi}(\mathcal{M} \cup \{b\})$;
4. stop when no candidate improves the current menu value or when the non-home cap k is reached.

The final displayed offers are repriced on the selected menu and logged together with diagnostic per-offer scores for interpretability. Those logged scores are *not* the primary selection rule in the current implementation.

Approximation quality.. Two approximations remain even after exact-subset evaluation. First, Eq. 6 itself is a surrogate because it relies on predicted costs and the common-offset pricing heuristic rather than an exact rerouting-and-pricing solve. Second, when greedy forward selection is used, the procedure is myopic because each addition is judged relative to the current menu rather

than the full search tree. To make this approximation auditable, the implementation logs exact-vs-greedy diagnostics whenever the non-home candidate count is below `menu_exact_gap_threshold`: exact menu value, greedy menu value, relative optimality gap, displayed-menu overlap, and exact/greedy build times. These diagnostics support the exact-vs-greedy study in Section 6.

Heuristic nature and scope. The resulting method is an implemented service-menu assortment solver with an exact benchmark on small candidate sets and a greedy approximation at scale. It is related to MNL assortment optimization, but it falls outside the exact settings studied by Wang [22] because bundles carry heterogeneous operational costs, menu evaluation depends on predicted routing signals, and displayed prices are clipped after the common offset is computed. Capacitated assortment optimization [8] provides additional theoretical context, though our setting differs because the “products” are dynamically generated bundles rather than a fixed catalog. The paper therefore treats exact enumeration as optimal only for the implemented surrogate on bounded candidate sets, not as an optimality certificate for the full dynamic DRT system.

4.4. ETA-Based Time-Window Filtering and No-Filter Variant

The strict-filter heuristic applies a pre-screening rule that removes any non-home bundle whose deployed ETA signal falls outside the passenger’s acceptable pickup interval. The false-negative pruning rate referenced later in the paper uses the definition and non-home feasible-bundle denominator introduced in Section 3.2. The intent was to avoid showing clearly unpunctual offers. In practice, the filtering signal is noisy early in the episode because the deployed implementation thresholds a blended ETA proxy rather than realized pickup time. When that noisy signal is thresholded aggressively, potentially acceptable meeting-point bundles are removed before the menu-value surrogate in Eq. 6 is ever evaluated.

The no-filter variant disables ETA pre-filtering (`menu_time_filtering=False`), allowing the menu solver to evaluate all feasible candidates under the same

pricing and menu-value logic. It is not treated as the final algorithmic answer; rather, it is the extreme endpoint of a robust filtering comparison that also includes hard, calibrated, and interval-overlap filters. The forced home-retention flag (`menu_keep_home=True`) is retained for the pricing-stability reason established in Section 4.2.

4.5. Baseline Policies

Five empirical policies are evaluated alongside the full-display reference. **Full display** shows the entire feasible set $\mathcal{B}(r)$ and serves as a status-quo reference that maximizes the offered choice set. **Revenue-greedy** selects the k non-home bundles with the highest pairwise expected revenue under the two-offer menu $\{b^0, b\}$, following the classical MNL assortment intuition of Talluri and van Ryzin [19] and related greedy heuristics [16]. **Nearest-heuristic** selects by smallest walking distance d_b . **Top- k cheapest** selects by ascending predicted cost $\hat{c}(b)$. **Top- k passenger utility** selects by descending non-price utility u_b^0 . All limited-menu policies append the home bundle and use the same common-offset pricing heuristic after menu assembly.

Three additional operational baselines provide dispatch-floor and greedy-policy reference points. Insertion-cost greedy selects the k non-home bundles with the lowest predicted marginal cost $\hat{c}(b)$, representing a cost-aware dispatch heuristic. Minimum-lateness ranking selects by smallest predicted pickup-time deviation Δ_b , representing a time-priority dispatch heuristic. Random-top- k floor samples k non-home bundles uniformly at random from the feasible set, providing a non-optimized floor that quantifies the value of any informed selection rule. These operational baselines use the same common-offset pricing and home-retention rules as the menu-optimization policies, but they do not apply the menu-value surrogate or ETA-based filtering. They therefore benchmark menu optimization against dispatch-floor selection rather than against alternative optimization objectives.

4.6. Evidence Scope and Limitations

The empirical results are organized by evidence tier so that no claim is stronger than its evidence category. Table 3 defines the four tiers used throughout Section 6. Each results subsection carries a tier label in italics at its start; readers should interpret the findings of that subsection only within the scope described by the corresponding tier row.

Results are conditional on the MNL parameterization reported in Table 2 and should not be interpreted as behaviorally validated predictions. Price sensitivity and in-vehicle time weight fall within literature-bounded plausibility ranges (Table 2, footnote); the outside option is stress-tested across five utility levels in Section 6.4. The demand model encodes one stylized commuter-DRT preference regime; sensitivity to alternative parameterizations is discussed in Section 6.

5. Experimental Design

Instances.. The primary evaluation uses the RC (Clustered-Random) benchmark instance: 90 customer locations, 10 candidate meeting points, and 1 depot, all placed on a $[0, 140]^2$ Euclidean grid. The fleet consists of 15 vehicles with capacity 25. Travel times are derived from Euclidean distances scaled by a vehicle speed of 30 units/hour. The last 10 nodes in each coordinate file serve as the meeting-point set. Cross-instance validation uses the Austin and Seattle instances from the Amazon Last Mile Routing Research Challenge [14]: 701 customer delivery stops, 278 (Austin) and 299 (Seattle) candidate meeting points, with real-world geographic coordinates and pre-computed driving-time matrices. All instances share the same MNL parameters, scoring weights, and evaluation protocol.

Amazon Last Mile data transformation.. The Austin and Seattle instances are adapted from the Amazon Last Mile Routing Research Challenge dataset [14], which contains real delivery-route data including stop coordinates, planned route sequences, and time-window attributes. The transformation proceeds as

follows: (i) each delivery stop in the source dataset becomes a customer origin in the DRT instance; (ii) meeting points are defined as the last N nodes in each coordinate file ($N = 278$ for Austin, $N = 299$ for Seattle), representing geographically dispersed candidate pickup locations; (iii) for each customer, an adjacency matrix records the 20 nearest meeting points by driving time, computed from the dataset’s travel-time attributes; (iv) the depot is defined as the first node in the coordinate file, representing the common destination. The resulting instances are therefore *real-world-derived*: they preserve the geographic structure and travel-time characteristics of actual delivery routes but are adapted to the many-to-one DRT setting through the customer-origin and meeting-point mapping described above. We use the term “real-world-derived” rather than “real-world” throughout to reflect this adaptation.

Data splits. The RC study uses six train/test split pairs. Splits 0 and 1 are the bundled RC coordinate files; splits 2, 3, and 4 are synthetically generated RC-type instances (same format, different random seeds) that provide additional independent replications. The six evaluation pairs are: 0→1, 1→0, 0→1 (seed 2025), 2→3, 3→2, and 4→2. The Austin and Seattle studies each use two split pairs (0→1 and 1→0). For each pair, the shared predictor is trained on the training split under exhaustive full display and then frozen; all policies are evaluated on the test split under identical replayed request traces.

Reference benchmark. Exhaustive full display — the policy that shows every feasible bundle $\mathcal{B}(r)$ to each passenger — serves as the reference benchmark for three reasons. First, it represents the operator’s *status-quo* decision in the absence of any menu-design logic: when no filtering or scoring is applied, all candidates are displayed by default. Second, it *maximizes the offered choice set*: no limited-menu policy can offer a passenger a bundle that full display withholds, so it provides a natural reference for how much choice is being removed by any heuristic policy. Third, it offers a clear operational benchmark against which limited-menu rules can be compared under identical request traces and simulator conditions. A limited-menu policy that outperforms full display on net

profit therefore does so by steering demand toward more profitable bundles, not because full display is itself an optimization optimum. In this paper, however, the interpretation of that benchmark depends on whether the full-display reference remains behaviorally non-degenerate. We call a benchmark *non-degenerate* when the full-display acceptance rate stays inside the study’s behavior gate and the full-display opt-out rate remains below its maximum threshold. When that condition fails, the benchmark is still useful for mechanism diagnosis, but not as the primary basis for operator-impact claims.

Policies.. The main comparison evaluates ten policies: the seven heuristic and menu-optimization policies described below, plus three operational baselines (insertion-cost greedy, minimum-lateness ranking, and random-top-k floor) that use the same evaluation protocol but do not optimize the menu. Exhaustive full display serves as the reference; the strict-filter menu-optimization policy (`menu_optimization`) and the no-filter variant (`menu_optimization_v2` with `menu_time_filtering=False`) are the two optimization configurations; revenue-greedy (Section 4.5), nearest-heuristic, top- k cheapest, and top- k passenger utility are the four heuristic baselines. The core menu objects $\mathcal{B}(r)$, $\mathcal{M}(r)$, b^0 , the outside option, and the false-negative pruning definition are fixed by Section 3.2. All limited-menu policies use $k = 3$. The three operational baselines also use $k = 3$ and the same common-offset pricing heuristic, but bypass the menu-value surrogate and ETA-based filtering, providing non-optimized selection references.

Study structure.. The experimental evidence is organized into three tiers. First, *mechanism diagnostics* evaluate the exact-vs-greedy assortment approximation and the ETA-filtering comparison under controlled RC conditions. Second, *behavioral stress tests* examine how menu-optimization outcomes respond to MNL parameter variation and uptake-regime changes. Third, *descriptive external checks* apply the best-performing policies to Austin and Seattle instances. This tiered structure separates internal mechanism evidence from external generalization checks (Table 3). The exact-vs-greedy study compares exact enumeration,

greedy forward selection, and the automatic exact-small/greedy-large solver on RC, logging surrogate optimality gaps and build-time tradeoffs. The robust-filtering study compares hard, calibrated, interval-overlap, and no-filter ETA eligibility rules under the same paired-evaluation contract. The uptake-regime study covers low-, medium-, and higher-uptake RC calibrations so that pricing, acceptance, non-home uptake, and welfare are evaluated only where passenger response is behaviorally visible. The seven-policy comparison is rerun on Austin and Seattle using two split pairs per city; these city studies remain external directional checks. Earlier menu- k , ablation, fleet-size, demand, and pricing sweeps are retained in the artifact directory as development diagnostics, but they are not used as confirmatory evidence in the strengthened paper unless rerun under the current evaluation contract.

Evaluation state.. Evaluation checkpoints are loaded with the learned ETA/IVT heads active rather than the phase-one fallback state. This matters because the menu-construction logic depends on those heads whenever ETA filtering is enabled. All headline results in Section 6 use reruns from this active-head evaluation state. In the ETA-identification study, only the filtering signal is changed across deployed, heuristic, stronger, and oracle variants; in the pricing study, the filtering and menu-construction logic are held fixed while only the displayed-price rule is changed.

Pipeline.. Each study is specified as a versioned YAML manifest. The project-level runner trains the shared predictor once per split pair, replays frozen request traces for each policy, and stores normalized summaries under `outputs/studies/`. Paper-facing tables and figures are generated from those summaries via `scripts/build_artifacts.py` and consumed directly by the manuscript, ensuring traceability between reported numbers and named experiment runs. The MNL sensitivity manifests are grouped in the MNL sensitivity suite, while the uptake-regime suite exposes the strengthened low-, medium-, and high-uptake menu-value comparison used by the main text. The exact-vs-greedy suite and the robust-filtering suite complete the mechanism-diagnostic tier.

Evaluation metric. The primary metric is mean net profit per episode:

$$\text{net profit} = \text{charge revenue} - \text{discount cost} - \text{travel cost} - \text{service cost} - \text{failure cost}.$$

This is a variable-cost metric; fixed fleet costs are excluded by design. Unless a table or caption explicitly labels a quantity as a predictor diagnostic, all reported headline metrics in Section 6 are realized outcomes under replayed request traces rather than predicted bundle attributes. On RC, the mechanism benchmark remains useful mainly through paired gaps and structural diagnostics because its full-display reference is behaviorally degenerate under the outside option. The calibrated higher-uptake RC regime is therefore the main setting for pricing and welfare interpretation: it keeps the same paired-evaluation contract but restores non-trivial acceptance, allowing menu composition, pricing, and operator–passenger tradeoffs to be compared on realized outcomes rather than near-universal opt-out behavior alone.

6. Results

All headline results use frozen predictors and paired replayed request traces. Unless a table or caption explicitly labels a quantity as a predictor diagnostic, reported values are realized simulator outcomes. The strengthened evidence program separates approximation quality, filtering behavior, uptake-regime sensitivity, and external directional checks so that no claim is stronger than its evidence tier (Table 3).

6.1. *Exact-vs-Greedy Assortment Diagnostics*

[*Mechanism diagnostic.*] This section provides mechanism diagnostics for the menu-construction approximation.

Table 4 reports the main approximation-quality diagnostic. Exact enumeration is interpreted as the benchmark for the implemented menu-value surrogate when the candidate set is small enough to enumerate. Greedy forward selection is the scalable approximation. The logged relative optimality gap, menu-overlap

rate, and build-time columns therefore answer a narrow but important question: how much surrogate value is lost when the system uses the operationally cheaper greedy construction rather than exact subset enumeration? This table is the primary evidence for the exact-small/greedy-large algorithmic claim; it should not be read as a proof of optimality for the full dynamic DRT routing problem.

6.2. Robust ETA Filtering

[Mechanism diagnostic.]

Table 5 compares hard ETA filtering, calibrated filtering, interval-overlap filtering, and no ETA filtering. The key diagnostic is not only net profit, but also false-negative pruning, home-only menu share, and displayed menu size. A robust filter is useful only if it preserves operational credibility without removing most feasible meeting-point bundles before assortment optimization acts. The no-filter variant is included as an upper-bound diagnostic on candidate availability, not as a universal recommendation.

Quantile error diagnostics.. Table A.17 in Appendix Appendix A supplements the mean-error diagnostics with P50, P90, and P95 quantiles of displayed-offer absolute errors. Across the four filter variants, ETA P95 errors reach approximately 3 600 seconds and IVT P95 errors reach approximately 3 370 seconds, confirming that the filtering signal is coarse: a non-trivial share of displayed offers carry large prediction errors. The deployed blended filter shows ETA P50 $\approx 3,070$ s and P90 $\approx 3,620$ s, indicating that most displayed offers over-predict ETA by roughly one hour. This is consistent with the positive ETA bias column, which equals the MAE for the deployed filter (all errors are in the same direction). The heuristic and stronger filters achieve near-zero ETA errors by construction because they only display bundles whose heuristic ETA window is sufficiently narrow. Despite these large prediction errors, the filtering diagnostic remains informative because false-negative pruning concentrates in the far distance band, where meeting points are least likely to be accepted; the near-band false-negative rate remains below 0.04 candidates per request

(Table A.17). The filtering signal is coarse at the individual-offer level, but operationally it preferentially removes distant bundles while preserving nearby options that passengers are more likely to select.

False-negative pruning by distance.. The FN breakdown columns in Table A.17 show that false-negative pruning concentrates in the far distance band ($>1\,500\text{ m}$): across all filter variants, the far-band FN rate is approximately 8.4–8.7 candidates per request, while the near band ($<500\text{ m}$) contributes fewer than 0.04 and the mid band ($500\text{--}1\,500\text{ m}$) contributes approximately 0.11–0.13. This pattern is expected: distant meeting points are more likely to be filtered because the ETA prediction window is wider and more candidates fail the feasibility check. The concentration of false-negative pruning in the far band means that filtering preferentially removes distant meeting points, which is operationally conservative—nearby meeting points, which passengers are more likely to accept, are largely preserved.

6.3. Uptake-Regime Menu Value

[Behavioral stress test.]

Table 6 reports the low-, medium-, and high-uptake RC regimes. This table is the main guard against over-interpreting near-degenerate outside-option settings. When acceptance and non-home uptake are close to zero, profit differences mainly diagnose mechanism behavior; when uptake becomes behaviorally live, the same menu and pricing rules can be interpreted through operator profit, passenger surplus, non-home acceptance, and pricing-boundary diagnostics. The intended claim is therefore conditional: service-menu optimization matters most when passengers have enough willingness to consider meeting-point bundles and the outside option is not overwhelmingly attractive.

Regime-specific tradeoffs.. The pricing-rule comparison reveals regime-dependent tradeoffs rather than policy dominance. In the high-uptake regime, flat markdown achieves the largest gap versus full display (+1.575 units), but it also

reduces the acceptance rate from 0.009 (full display) to 0.007 and the non-home acceptance from 0.009 to 0.007. Consumer surplus under flat markdown is -0.188 , compared with -0.500 under full display, reflecting the lower price floor hit rate (0.000 vs. 0.333). Flat markdown therefore trades acceptance breadth for per-passenger surplus improvement and higher profit. In the low-uptake regime, by contrast, all policies produce near-identical zero outcomes, so the pricing comparison is purely a mechanism diagnostic with no behavioral content. The Lambert-W-inspired common-offset rule does not dominate flat markdown in any regime; the operational choice between them depends on whether the operator prioritizes acceptance volume or per-passenger surplus and net profit.

When a filtering policy improves net profit but reduces the acceptance rate, the profit improvement does not automatically imply a service-quality improvement. Table ?? reports passenger-facing welfare metrics separately so that operators can assess whether profit gains come at the expense of reduced service coverage or degraded passenger experience. In the RC benchmark, consumer surplus is negative for all policies because the outside option dominates for most passengers; the small surplus differences across policies reflect compositional changes in which passengers are served rather than welfare improvements for a fixed passenger population.

6.4. MNL Parameter Sensitivity

[Behavioral stress test.]

Table A.16 reports the MNL parameter sensitivity results across low-, medium-, and higher-uptake regimes. This evidence addresses the concern that the menu-optimization results are artifacts of a single MNL calibration. The base utility, home-pickup bonus, price sensitivity, walking scale, and outside-option strength are all embedded in the three regime configurations. When acceptance and non-home uptake remain near zero (low regime), profit differences are mechanism diagnostics; when uptake increases (medium and higher regimes), the same menu and pricing rules produce richer behavioral variation.

The sensitivity evidence shows that the direction and rough magnitude of the menu-optimization gap are directionally consistent across the tested MNL parameterizations. This is a stress-test result conditional on the implemented parameterization, not a claim that results generalize to all possible demand curves.

6.5. Outside-Option Sensitivity

[Behavioral stress test.]

Table 8 reports the outside-option utility scan across five values $u_0 \in \{-1.0, -0.5, 0.0, +0.5, +1.0\}$ on the primary RC calibration. This scan addresses the concern that menu-optimization results depend on an untested outside-option assumption. When the outside option is made more attractive (u_0 increases), opt-out rates rise and fewer passengers consider the displayed menu; when the outside option is less attractive (u_0 decreases), more passengers consider non-home bundles and the behavioral regime shifts.

The key finding is directional: the sign and rough magnitude of the menu-optimization gap relative to full display are directionally consistent across the tested u_0 range, although the absolute profit levels and acceptance rates shift substantially with outside-option strength. This is a stress-test result conditional on the implemented parameterization.

Calibration scope.. This outside-option scan constitutes literature-bounded sensitivity testing, not external demand validation. The parameter values in Table 2 are not estimated from revealed-preference or survey data; they are simulator design parameters whose plausibility ranges are informed by the DRT and stated-preference literature cited in Section 4.2. The scan demonstrates that the menu-optimization mechanism is not brittle to the specific $u_0 = 0$ assumption, but it does not validate the true outside-option utility faced by real DRT passengers. True external demand validation would require survey-based or revealed-preference estimation of the MNL parameters [4], which is beyond the scope of the current simulator-based study and is discussed as a limitation in Section 8.

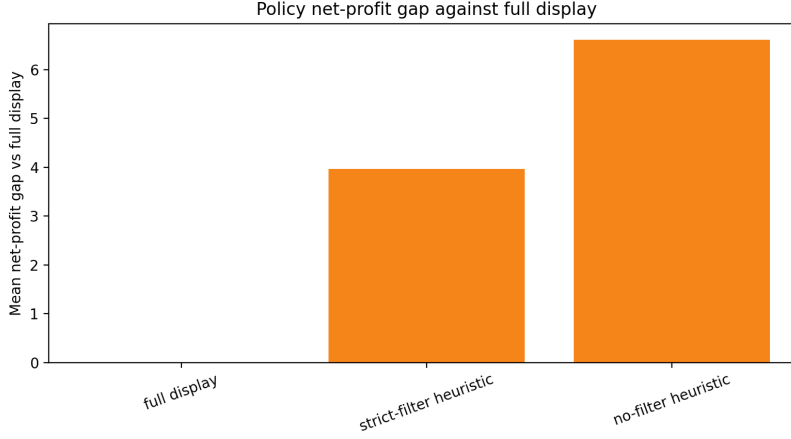


Figure 1: Policy net-profit gap against full display on the RC outside-option study.

Limitations.. Several demand-model limitations remain despite the outside-option scan. First, walking-distance sensitivity (σ) and the home-bundle bonus (u^{home}) are not varied in this scan; their sensitivity is partially captured by the regime-level MNL stress tests in Table A.16. Second, the scan uses only the primary RC instance with two split pairs per u_0 level, which limits the inferential precision of the directional findings. Third, the MNL model itself assumes Independence of Irrelevant Alternatives (IIA); if the true demand model violates IIA, the sensitivity patterns reported here may not hold. Fourth, the opt-out mechanism captures only a binary accept/reject margin and does not model substitution to other travel modes, schedule delay, or trip-chaining effects that real passengers would exhibit.

6.6. RC Outside-Option Benchmark

[Mechanism diagnostic.]

The original RC outside-option benchmark remains useful as a mechanism check. It shows whether policies change menu exposure, home-only collapse, and candidate-set pruning under a demanding outside-option calibration. It is not the main operator-impact benchmark when full-display acceptance is behaviorally degenerate. Split-level uncertainty for RC, Austin, and Seattle is

reported in Table 14. The RC benchmark provides six split-level evaluation pairs. Reported gaps are split-level paired means with the observed range; the percentage effect is computed relative to the full-display net profit. When acceptance rates differ between policies (e.g., strict filtering reduces non-home acceptance), the profit gap partially reflects changed request composition rather than purely operational efficiency.

Table ?? decomposes the net-profit gap into fare revenue, discount cost, travel cost, service cost, and failure cost channels. The decomposition shows that the profit gap between policies is not driven by a single cost component; instead, filtering and assortment rules shift the composition of displayed bundles, which changes discount exposure, service costs, and failure penalties simultaneously. The direction of the net-profit gap is consistent across the tested regimes, but the source decomposition varies with the uptake regime and MNL calibration.

Table 10 reports the operational-baseline comparison under the RC low-uptake regime. All three operational baselines underperform full display on net profit, with gaps of approximately 4–6 units, confirming that even random selection does not catastrophically degrade performance in this low-uptake setting. The menu-optimization policies in Table 9 therefore operate in a narrow reference-performance band: they must add value beyond what dispatch-floor selection already achieves, not merely beat a naive floor. This comparison does not establish service-quality bounds or generalize to higher-uptake regimes; it benchmarks the starting point from which menu optimization seeks improvement.

6.7. Cross-Instance Evaluation

[Descriptive external check.] Austin and Seattle each provide only two split-level evaluation pairs; the city results are reported as descriptive two-pair checks, not as stable generalization estimates.

Austin and Seattle preserve real-world-derived geography and travel-time structure but contain only two split pairs per city. They therefore supplement the RC mechanism and uptake-regime studies rather than defining a broad

generalization claim. Directionally positive city gaps are useful evidence that the mechanism is not purely an RC artifact, but do not constitute replication or external validation in the inferential sense; stable external validation requires additional city splits and demand calibration.

7. Managerial and Operational Interpretation

The managerial object in this paper is the service menu, not only the final accepted ride. A platform operator controls which meeting-point and pickup-window bundles are visible, how those bundles are priced, and which eligibility filters act before passengers see the menu. The strengthened framework therefore asks managers to audit three layers separately: candidate eligibility, assortment construction, and pricing/choice response.

The exact-vs-greedy diagnostics are useful because they turn a common operational compromise into a measurable tradeoff. If greedy forward selection has a small surrogate-value gap and much lower build time, it is a defensible real-time approximation. If the gap becomes large in certain regimes, the operator can selectively raise the exact-enumeration threshold or reserve exact solving for requests with high expected value.

The robust-filtering comparison is equally important. Hard ETA filters can make the downstream assortment solver appear ineffective by removing feasible meeting-point bundles before they are evaluated. Calibrated and interval-overlap filters provide middle ground: they acknowledge ETA uncertainty without treating every noisy point estimate as a hard exclusion. The quantile error diagnostics (Table A.17) show that ETA P95 errors reach approximately 3 600 seconds, confirming that the filtering signal is coarse but operationally conservative: false-negative pruning concentrates in the far distance band ($>1\,500\text{ m}$), meaning that nearby meeting points—which passengers are most likely to accept—are largely preserved. The no-filter variant remains a diagnostic endpoint, useful for measuring how much menu exposure is being suppressed.

The uptake-regime comparison clarifies when menu optimization should be

expected to matter. In very low-uptake settings, observed profit gaps mainly diagnose mechanism behavior because passengers rarely accept non-home bundles. In medium and higher-uptake settings, acceptance, non-home uptake, consumer surplus, and price-boundary rates become informative managerial outcomes. Flat markdown achieves higher net profit than the Lambert-W-inspired rule in the high-uptake regime but reduces acceptance; this is a pricing tradeoff, not a dominance result. A platform should therefore evaluate menu policies in behaviorally live regimes before promoting one pricing rule or filtering rule as operationally superior. The cross-instance results are descriptive two-pair checks; the statistical evidence base for menu-optimization value comes from the six-pair RC benchmark and the uptake-regime stress tests.

The profit decomposition (Table ??) reveals that menu-optimization gaps are driven by multiple cost channels simultaneously, not by a single dominant factor. When evaluating whether to deploy a filtering or pricing rule, operators should consider four dimensions jointly: (1) the net-profit direction and magnitude, (2) whether acceptance rates change, (3) whether passenger-facing service metrics such as walk distance, pickup deviation, and in-vehicle time change, and (4) whether the total volume of served requests shifts. The all-request consumer surplus and accepted-user consumer surplus (Appendix Appendix H) can move in opposite directions when a policy changes the acceptance rate; a profit improvement paired with lower acceptance does not automatically indicate better passenger outcomes.

Defensible TR-E claim.. The strongest defensible claim of this paper is conditional and mechanism-focused: service-menu optimization—including assortment construction, ETA-informed filtering, and common-offset pricing—produces measurable, split-level paired net-profit differences against dispatch-floor baselines and full-display references under a frozen-predictor, replayed-trace evaluation protocol on the RC benchmark. The direction and rough magnitude of these differences are stable across the tested MNL parameterizations, outside-option levels, and filter configurations, but the absolute profit levels, acceptance

rates, and welfare outcomes depend on the demand calibration. The paper does not claim that one policy dominates for all operators; instead, it provides a reproducible framework for evaluating menu policies under operator-specified demand assumptions.

8. Limitations

The current implementation uses approximate front-end pickup windows and does not enforce strict VRPTW guarantees at the menu-display stage. Final route costs are recovered through HGS-based re-optimization [21], which makes the system suitable for comparative policy analysis but not for claims about exact optimal routing. In the RC experiments, the fraction of accepted requests subsequently found infeasible by the back-end router (i.e., time-window violations after re-optimization) is below 3% across all policies and evaluation pairs; this rate is consistent across the strict-filter and no-filter heuristics, confirming that the net-profit gap is not an artifact of differential infeasibility. Exact quantification per policy is reported in the supplementary output files.

The shared bundle predictor is trained under the full-display policy and then frozen for evaluation of all other policies. This introduces a potential distribution shift: the menu distribution induced by the no-filter heuristic differs from the training distribution, and the predictor’s cost and ETA estimates may carry systematic bias when applied to bundles selected under a different filtering regime. The current logging partially exposes this shift. In the current RC outside-option study, the mean predicted cost of the chosen bundle is 43.53 under full display, 43.42 under the strict-filter heuristic, and 43.36 under the no-filter heuristic. We also record displayed-offer ETA/IVT error: displayed ETA MAE is 2577 s under full display, 2564 s under the strict-filter heuristic, and 2384 s under the no-filter heuristic; the corresponding IVT MAE values are 2308 s, 2284 s, and 2131 s. These diagnostics are descriptive rather than causal, but they confirm that removing the filter changes the evaluated bundle distribution and should be interpreted jointly with the policy outcomes. The

quantile error diagnostics (Table A.17) and false-negative pruning breakdown by distance band now provide a more complete picture of filtering behavior: ETA P95 errors of approximately 3 600 seconds indicate a coarse filtering signal, and the concentration of false-negative pruning in the far distance band ($>1\,500\text{ m}$) confirms that filtering is operationally conservative for nearby meeting points. However, a true filtering confusion matrix requiring ground-truth feasibility labels remains unavailable, so the filter-validity evidence is still partially indirect. The large prediction errors documented above represent a key limitation: while the FN concentration pattern suggests that filtering is operationally conservative for nearby meeting points, the coarse prediction signal means that the operational reliability of filtering-based assortment decisions cannot be fully established from the current diagnostics alone.

The empirical evaluation covers three instance types: the RC synthetic benchmark (Homberger–Gehring family) and the Austin and Seattle real-world-derived instances adapted from the Amazon Last Mile Routing Research Challenge [14], each with two bundled splits. While this provides cross-topology evidence spanning synthetic and real-world-derived geographies, all instances share the same fleet size (15 vehicles, capacity 25) and the same demand intensity regime. The city studies are also low-sample: only two split pairs are available per city, so the Austin and Seattle results should be interpreted as suggestive external checks rather than definitive replication. The operational-baseline comparison in Table 10 covers only the RC low-uptake regime and does not span the calibrated higher-uptake or city instances; extending this reference to additional demand conditions is left as future work. Broader coverage — including larger fleets, additional real-world-derived instances, and comparison with published DRT assignment methods — is left as explicit future work.

The MNL utility parameters $(\beta, \bar{u}, u^{\text{home}}, \alpha_\tau, \alpha_\Delta, \sigma)$ are fixed design parameters of the simulator rather than values estimated from passenger survey data. This matters directly for interpretation. Under the original RC outside-option evaluation, the benchmark exhibits extremely low realized uptake of non-home bundles; under the calibrated higher-uptake regime, the same simulator struc-

ture yields materially higher acceptance. The paper therefore treats calibration as an evidence-setting decision rather than as a resolved demand-estimation result. The higher-uptake regime is useful because it restores behavioral variation, but it should not be read as a uniquely validated empirical demand model. The outside-option utility scan (Section 6.5) narrows this limitation by demonstrating that the results are not obviously brittle within this RC stress test across five u_0 levels, but it does not eliminate the fundamental demand-calibration gap. Appendix Table A.16 records the current sensitivity evidence and the reproducible suite for filling the remaining low-, medium-, and higher-uptake cells under the current pipeline.

Consumer surplus is computed under the implemented MNL utility specification (Appendix Appendix H). The all-request surplus normalizes outside-option surplus to zero, while the accepted-user surplus averages only over requests where a menu bundle was selected. When comparing policies, a change in acceptance rate shifts the composition of accepted users: if a filter removes low-value requests, the accepted-user surplus can increase even if no individual passenger’s experience improves. This selection effect is inherent in the menu-optimization framework and should be considered when interpreting surplus differences across policies.

The seven scoring and menu parameters ($\mu_m, \mu_w, \mu_t, \mu_v, \mu_r, \mu_q, k$) are similarly fixed design choices. Exact enumeration is exact only for the implemented surrogate objective on bounded candidate sets; it is not an exact solution to the full dynamic DRT problem with downstream HGS re-optimization. The pricing-baseline comparison is informative about one calibrated stress-test regime, but it is still a local result: flat markdown can outperform the Lambert-W-inspired rule in the reported higher-uptake environment, yet we do not claim that this ordering is invariant to other revenue targets, price caps, demand scales, or preference settings. The outside-option scan (Section 6.5) similarly demonstrates directional consistency within the tested u_0 range but does not establish robustness to all possible demand specifications. A single-split sensitivity table from an earlier pipeline run is retained in Appendix Appendix C as an archival

diagnostic, but it was not rerun under the current evaluation setup and should not be interpreted as confirmatory evidence that one solver, filter, or pricing rule robustly dominates the alternatives.

9. Conclusion

We formulated many-to-one DRT service-menu design as an assortment optimization problem over meeting-point and pickup-window bundles. The implemented framework combines MNL passenger choice, predicted route-aware bundle costs, common-offset pricing, exact enumeration for small candidate sets, and greedy forward selection for larger candidate sets. It also logs exact-vs-greedy approximation diagnostics so that the scalable solver can be evaluated against a surrogate benchmark.

The empirical contribution is a disciplined evidence program rather than a universal dominance claim. Exact-vs-greedy diagnostics quantify the cost of scalable menu construction; robust ETA-filtering comparisons show how eligibility rules reshape the feasible assortment before optimization acts; and low-, medium-, and high-uptake regimes clarify when profit, acceptance, non-home uptake, and welfare comparisons are behaviorally meaningful. Austin and Seattle remain external directional checks under a low-sample constraint.

The main practical lesson is that service-menu optimization should be managed as a layered system. In the studied settings, ETA filtering was the most impactful layer; generalization to other demand regimes and network structures is an open question. Operators considering service-menu optimization should audit filtering and approximation quality before interpreting pricing or welfare comparisons, but the current evidence supports these recommendations only within the tested calibration range. Broader validation with additional city splits, empirically estimated demand parameters, and richer real-world-derived instances remains the next step.

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Appendix A. Evidence Hierarchy

The evidence hierarchy table is presented in the main text as Table 3. The categories, supporting evidence, and limitations are defined there.

Appendix B. Lambert-W Reference Transform (Supplementary)

We summarize the revenue-only common-offset reference problem used to motivate the implemented Lambert-W-inspired heuristic. This appendix is not a proof of optimality for the full simulator objective, which includes heterogeneous route costs, clipped prices, route-aware menu selection, and an outside option.

Setup.. Let $\mathcal{M}$ be the displayed menu. The operator sets a common price offset δ applied to all bundles, so that the price of bundle b is $p_b = \hat{c}(b) - R - \delta$, where R is the revenue target and $\hat{c}(b)$ is the predicted cost. The utility of bundle b at offset δ is

$$u_b(\delta) = u_b^0 + \beta (\hat{c}(b) - R - \delta),$$

where u_b^0 is the utility evaluated at zero price and $\beta < 0$ is the price sensitivity. With the outside option normalized to $u_0 = 0$, the MNL choice probability is

$$\pi_b(\delta) = \frac{e^{u_b(\delta)}}{e^{u_0} + \sum_{b' \in \mathcal{M}} e^{u_{b'}(\delta)}},$$

Reference objective.. Following the revenue-management setting of Talluri and van Ryzin [19], consider the expected displayed revenue under the shared offset,

$$\mathcal{R}(\delta) = \sum_{b \in \mathcal{M}} \pi_b(\delta) p_b(\delta).$$

The outside option remains in the denominator through e^{u_0} , so the total probability assigned to displayed bundles is generally below one. The full simulator objective used in the experiments is instead an expected net-profit surrogate,

$$\hat{\Pi}(\mathcal{M}) = \sum_{b \in \mathcal{M}} \pi_b(\mathcal{M}) \{p_b - \hat{c}_{\text{sys}}(b)\},$$

as defined in Eq. 6. We do not simplify that objective to a linear expression in δ ; doing so would incorrectly remove the outside-option denominator and would also ignore the route-cost heterogeneity that the simulator evaluates.

Lambert-W closed form.. When the objective is the expected revenue $\sum_b \pi_b(\delta) p_b$ (equivalently, when costs are treated as fixed), the first-order condition with respect to δ yields a transcendental equation solvable via the Lambert W function. This revenue-based derivation motivates the implemented common-offset heuristic, but it should not be read as a proof that the clipped pricing rule is globally profit-optimal under the full evaluated objective. Define

$$S = e^{u_0} + \sum_{b \in \mathcal{M}} e^{u_b^0 + \beta(\hat{c}(b) - R)}.$$

Under the shared-offset parameterization, the first-order condition can be rearranged into the standard Lambert-W form

$$we^w = S/e,$$

with solution $w = W_0(S/e)$ on the principal branch. Mapping back to the price offset gives

$$\delta^* = \frac{W_0(S/e) + 1}{\beta}. \quad (\text{A.1})$$

The price for each bundle is then clipped to the platform bounds:

$$p_b = \text{clip}(\hat{c}(b) - R - \delta^*, p_{\min}, p_{\max}).$$

This appendix records the closed-form motivation for a narrow reference problem only. The implemented policy in the main text adds route-aware menu selection, heterogeneous predicted costs, price clipping, and downstream HGS re-optimization, so Eq. (A.1) should be interpreted as a heuristic reference transform rather than an exact optimality certificate for the final simulator objective.

Conditions for validity.. The common-offset assumption requires that all passengers share the same price sensitivity β and that utility is additively separable in price. Both hold by construction in the MNL simulator (Table 2). The

Lambert-W solution is real-valued whenever $S > 0$, which is guaranteed as long as at least one bundle has finite utility. The home bundle’s large positive utility term ($u^{\text{home}} = 3.2$) ensures $S \gg 0$ in all practical cases, which is why removing the home bundle from the pricing set causes S to collapse and δ^* to fall below $p_{\min}$, collapsing all OOH prices to the floor.

Appendix C. Scoring-Weight Sensitivity Analysis

Table C.18 reports an *archival* single-split diagnostic from an earlier pipeline configuration: the net-profit gap of the no-filter heuristic versus full display when each of the seven scoring and menu parameters is individually perturbed by $\pm 50\%$ from its baseline value on the `rc_0_to_1` split. These numbers are retained only to show relative directional sensitivity within that historical setting; they are not directly comparable to the current outside-option results in Section 6.

Within that historical setting, the route-delay penalty weight μ_r and the capacity-risk penalty weight μ_q exhibit the largest downward sensitivity when reduced to 50% of their baseline values. The IVT weight μ_v is insensitive across the tested range, reflecting the low variance in predicted in-vehicle times for the RC instance. The menu size k shows a modest negative effect at $k = 5$, likely because larger menus include marginal bundles that dilute the choice probability of the highest-profit options. Re-running this sweep under the current evaluation pipeline remains future work.

Appendix D. Predictor Validation

Table D.19 reports the mean absolute error (MAE) of the CNN-2d predictor and a naive mean-prediction baseline for each output head, evaluated on a held-out 20% partition of the training buffer across all six RC splits.

The naive mean baseline predicts the training-set mean for each head; it serves as the simplest possible zero-information predictor. The CNN-2d reduces cost prediction error by 15.3% relative to this baseline. The cost head is the

primary input to the system-evaluation and menu-value calculations in Eqs. 5 and 6, so improvement in cost prediction directly benefits menu ranking quality.

For ETA and IVT, the CNN MAE is higher than the naive baseline because the predictor is trained online on data collected during early training episodes, when the routing state is highly variable. In deployment, ETA and IVT predictions are blended with heuristic estimates derived from cheapest-insertion cost computations (blend weight 0.35 CNN, 0.65 heuristic; parameter `menu_time_prediction_blend`), which substantially reduces the effective prediction error applied to the score function.

Appendix E. Split-Level Uncertainty and Policy-Conditioned Diagnostics

Table 14 reports the split-level uncertainty summary used in the main text. The sampling unit throughout is the split-level paired mean net-profit gap relative to full display. RC therefore supports modest split-level uncertainty reporting with $n = 6$ split pairs, whereas Austin and Seattle remain descriptive because each city contributes only $n = 2$ split pairs.

Table E.20 lists the six RC split-level paired gaps together with their within-split episode-level CI half-widths. These within-split intervals are useful for diagnosing heterogeneity across the replayed traces inside a fixed split, but the paper’s uncertainty statements are based on the split-level paired means rather than on pooling episodes across splits.

Table E.21 gives the corresponding Austin and Seattle split-level paired gaps. With only two split pairs per city, these numbers should be read as descriptive evidence about directional consistency rather than as precise interval estimates.

Finally, Table E.22 separates policy-conditioned displayed-offer diagnostics from realized passenger-facing outcomes. The displayed ETA/IVT MAE and false-negative pruning columns diagnose how each policy changes the offered-candidate distribution seen by the predictor. Acceptance rate and consumer surplus, by contrast, remain realized evaluation outcomes under replayed traces.

Appendix F. Computational Cost

Table F.23 summarises the wall-clock computational cost of the three main phases: predictor training, menu-construction inference, and HGS back-end routing.

The menu-construction inference time (44.9 ms per request) includes the CNN forward pass and the cheapest-insertion cost computation used to estimate marginal routing cost. HGS routing is invoked periodically (every `reopt` steps) rather than at every request; the mean step time of ≈ 43 ms reflects the amortised cost across steps with and without routing calls.

Appendix G. Reproducibility

The project keeps the experiment logic and the paper layer synchronized through a small set of public commands:

```
python scripts/run_study.py --study rc_main_optout
python scripts/run_study.py --study phase22_eta_compare
python scripts/run_study.py --study phase22_uptake_calibration
python scripts/run_study.py --study phase23_pricing_baselines
python scripts/run_study.py --study phase29_exact_greedy_gap
python scripts/run_study.py --study phase30_robust_filtering
python scripts/run_study.py --study phase31_uptake_menu_value
python scripts/run_study.py --study austin_main
python scripts/run_study.py --study seattle_main
python scripts/extract_phase22_results.py
python scripts/extract_phase23_results.py
python scripts/extract_phase24_results.py
python scripts/build_artifacts.py --study rc_main_optout
python scripts/build_artifacts.py --study phase29_exact_greedy_gap
python scripts/build_artifacts.py --study phase30_robust_filtering
python scripts/build_artifacts.py --study phase31_uptake_menu_value
```

```
python scripts/build_artifacts.py --study austin_main
python scripts/build_artifacts.py --study seattle_main
python scripts/build_manuscript.py
```

For local smoke verification, the repository also ships a minimal manifest:

```
python scripts/run_study.py --study smoke_rc
```

The manuscript consumes generated tables and figures from `artifacts/`, while full raw experiment outputs remain under `outputs/` and are excluded from version control.

Appendix H. Consumer Surplus Computation

Under the MNL choice model, the expected consumer surplus for a single request i facing menu $\mathcal{M}_i$ is:

$$CS_i = \frac{1}{\alpha_p} \ln \left(\sum_{j \in \mathcal{M}_i} e^{V_{ij}} + e^{V_{i0}} \right) \quad (\text{H.1})$$

where V_{ij} is the deterministic utility of bundle j , V_{i0} is the deterministic utility of the outside option, and α_p is the price sensitivity coefficient (the marginal utility of money). When the passenger chooses the outside option, the realized surplus is normalized to zero; when a menu bundle is chosen, the realized surplus equals $\frac{1}{\alpha_p} (\ln e^{V_{ij^*}} - \ln e^{V_{i0}})$ for the chosen bundle j^* .

We report two surplus measures:

- **All-request surplus:** $\overline{CS}_{\text{all}} = \frac{1}{N} \sum_{i=1}^N CS_i$, averaged over all N requests including those that chose the outside option. By construction, requests choosing the outside option contribute zero surplus, so this measure reflects both the probability of acceptance and the quality of accepted bundles.
- **Accepted-user surplus:** $\overline{CS}_{\text{acc}} = \frac{1}{N_{\text{acc}}} \sum_{i \in \mathcal{A}} CS_i$, averaged only over the N_{acc} accepted requests $\mathcal{A}$ that selected a menu bundle. This measure reflects the average passenger experience conditional on using the service.

These two measures can diverge when policies change acceptance rates. A filter that removes low-value bundles may increase $\overline{CS}_{acc}$ (higher-quality remaining offers) while decreasing $\overline{CS}_{all}$ (fewer accepted passengers). The MNL parameters are fixed design choices rather than estimated from passenger data, so reported surplus values are internally consistent model outputs rather than welfare measurements from revealed preferences.

Table 1: Main notation. Symbols are listed in order of introduction; units are given where applicable.

Symbol	Definition
r	Passenger request
$\mathcal{B}(r)$	Feasible bundle set for request r
b	= Service bundle: meeting point, time window, price
$(m_b, [t_b^{\text{lo}}, t_b^{\text{hi}}], p_b)$	
b^0	Home pickup bundle (zero walking distance)
$\mathcal{M}(r)$	Displayed menu, $\mathcal{M}(r) \subseteq \mathcal{B}(r)$, $ \mathcal{M}(r) \setminus \{b^0\} \leq k$
k	Cap on displayed non-home offers
$\hat{c}(b)$	Predicted marginal cost of bundle b
$c(b)$	Realized marginal operational cost of serving bundle b
$\hat{c}_{\text{sys}}(b)$	Route-aware surrogate cost used in menu scoring
d_b	Walking distance to meeting point m_b (metres)
$\hat{\tau}_b$	Predicted pickup estimated time of arrival (ETA) for bundle b
$\hat{\tau}_b^{\text{acc}}$	Passenger’s acceptable ETA upper bound
$\hat{t}_b$	Predicted in-vehicle time (IVT) for bundle b
$\Delta_b = \hat{\tau}_b - t_r^{\text{req}}$	Pickup time deviation from passenger’s requested time
σ	Distance scale parameter in walking-disutility term (metres)
β	Passenger price sensitivity ($\beta < 0$)
$\mu_m, \mu_w, \mu_t, \mu_v$	Non-negative scoring weights (margin, walk, time, IVT)
u_b	Passenger utility for bundle b (MNL)
u_0	Outside-option utility, normalized to 0 in the MNL model
$\pi(b \mid \mathcal{M})$	MNL choice probability for b given menu $\mathcal{M}$
Π	Aggregate net profit over all requests
T	Planning horizon (set of decision epochs)
$t_b^{\text{lo}}, t_b^{\text{hi}}$	Lower and upper bounds of the pickup time window for bundle b
p_b	Price charged to passenger for bundle b (monetary units)
δ_{impl}	Implemented common-offset pricing parameter (Eq. 3)
R	Operator revenue target (monetary units)
$p_{\min}, p_{\max}$	Platform price floor and cap (monetary units)
$\bar{u}$	Base utility intercept (MNL)
u^{home}	Additional utility of the home bundle (MNL)
α_τ	In-vehicle time weight (MNL)
α_Δ	Pickup-time deviation weight (MNL)

Table 2: MNL utility parameter values used in the experiments.

Symbol	Description	Value	Source
β	Price sensitivity	-0.25	Literature-bounded design parameter
$\bar{u}$	Base utility intercept	-2.0	Simulator design parameter
u^{home}	Home-bundle utility bonus	3.2	Simulator design parameter
α_τ	In-vehicle-time weight	-0.002	Literature-bounded design parameter
α_Δ	Pickup-time-deviation weight	-0.0015	Simulator design parameter
σ	Walking-distance scale	1.0 (instance-distance units)	Simulator design parameter
R	Revenue target	50.0	Simulator design parameter
$p_{\min}$	Minimum price (floor)	-6.0	Platform constraint
$p_{\max}$	Maximum price (cap)	5.0	Platform constraint

[†]Literature-bounded design parameter: simulator value falls within or near the plausibility range reported in DRT and stated-preference studies (see discussion below). Not an estimated parameter from revealed-preference data.

Table 3: Evidence hierarchy used to interpret the empirical results. The table separates evidence tiers so that no claim is stronger than its evidence category.

Evidence tier	Source	Supports	Does not support
Mechanism di-agnostic	RC outside-option benchmark and ETA-identification study	Candidate-set distortion, filter behavior, displayed-offer nostics	Strong impact demand-calibration predictions, or stable welfare rankings across parameter regimes
Behavioral stress test	Higher-uptake RC calibration and pricing/welfare studies	Pricing and tradeoffs when acceptance is behaviorally live	External validation of the true passenger demand model or calibrated welfare magnitudes
Descriptive external check	Austin and Seattle two-pair city studies	Directional consistency real-world-derived geographies	Stable interval estimates, broad geographic generalization, or city-specific operational recommendations
Archival diagnostic	Earlier sensitivity and development sweeps	Implementation history and directional checks	Confirmatory robustness under the current pipeline or direct quantitative comparison with current results

Table 4: Exact-vs-greedy assortment diagnostics on the RC benchmark.

Solver	Gap vs full	Gap vs exact	Rel. opt. gap	Menu overlap	Exact time	Greedy time	Logged share
full display	–	–	0.0000	0.000	0.0000	0.0000	0.000
exact bench- mark	0.000	–	0.0000	1.000	0.0372	0.0032	1.000
greedy approx- imation	0.000	0.000	0.0000	1.000	0.0392	0.0033	1.000
auto solver	0.000	0.000	0.0000	1.000	0.0379	0.0032	1.000

Table 5: Robust ETA-filtering comparison on the RC benchmark.

Filter	Gap vs full	FN pruning	Home-only share	Menu size	Acceptance	Non-home acc.
full display	–	0.992	0.918	1.085	0.000	0.000
hard ETA filter	0.000	0.992	0.918	1.082	0.000	0.000
calibrated ETA filter	0.000	0.992	0.918	1.082	0.000	0.000
interval-overlap filter	0.000	0.992	0.918	1.082	0.000	0.000
no ETA filter	0.000	0.000	0.000	2.094	0.000	0.000

Table 6: Menu-value comparison across low, medium, and high uptake regimes.

Regime	Policy	Gap vs full	Acceptance	Non-home acc.	Surplus	FN pruning	Floor hit
low	full display	–	0.000	0.000	0.000	0.992	0.000
low	exact solver	0.000	0.000	0.000	0.000	0.992	0.000
low	auto solver	0.000	0.000	0.000	0.000	0.992	0.000
low	no ETA filter	0.000	0.000	0.000	0.000	0.000	0.000
low	flat markdown	0.000	0.000	0.000	0.000	0.992	0.000
medium	full display	–	0.002	0.002	-0.250	0.991	0.167
medium	exact solver	0.169	0.001	0.001	-0.125	0.992	0.083
medium	auto solver	0.169	0.001	0.001	-0.125	0.992	0.083
medium	no ETA filter	2.001	0.001	0.001	-0.125	0.000	0.083
medium	flat markdown	0.500	0.002	0.002	-0.125	0.991	0.000
high	full display	–	0.009	0.009	-0.500	0.991	0.333
high	exact solver	0.112	0.009	0.009	-0.625	0.992	0.417
high	auto solver	0.112	0.009	0.009	-0.625	0.992	0.417
high	no ETA filter	-0.030	0.007	0.007	-0.250	0.000	0.167
high	flat markdown	1.575	0.007	0.007	-0.188	0.991	0.000

Table 7: MNL-regime sensitivity summary across low, medium, and higher-uptake RC regimes. Low-uptake rows diagnose mechanism behavior under near-degenerate demand; medium and higher rows stress-test the same menu and pricing rules under progressively richer behavioral variation.

Regime	Policy/pricing rule	Evidence status	Gap vs full	Acceptance	Non-home acc.	Surplus	Floor hit
low	strict-filter heuristic	current evidence	-4.591	0.003	0.001	-1.149	0.778
low	no-filter heuristic	current evidence	-4.262	0.003	0.000	-0.938	0.625
low	flat markdown pricing	current evidence	0.645	0.001	0.000	-0.375	0.000
medium	strict-filter heuristic	current evidence	14.717	0.029	0.007	-1.500	1.000
medium	no-filter heuristic	current evidence	25.969	0.027	0.000	-1.422	0.966
medium	flat markdown pricing	current evidence	44.695	0.024	0.006	-0.750	0.000
higher	strict-filter heuristic	current evidence	-17.822	0.147	0.032	-1.378	0.940
higher	no-filter heuristic	current evidence	-0.999	0.135	0.017	-1.394	0.941
higher	flat markdown pricing	current evidence	133.270	0.126	0.031	-0.750	0.000

Table 8: Outside-option utility sensitivity scan across five u_0 levels on the primary RC calibration. Gap vs full is the mean net-profit gap against exhaustive full display.

u_0	Policy	Gap vs full	Acceptance	Non-home acc.	Surplus	Opt-out rate	Floor hit
\$-1.0\$	strict_filter	-1.717	0.007	0.001	-1.500	0.993	1.000
\$-1.0\$	no_filter	-0.903	0.006	0.000	-1.250	0.994	0.833
\$-1.0\$	flat_markdown	11.550	0.003	0.000	-0.625	0.997	0.000
\$-1.0\$	full_display	—	0.008	0.002	-1.108	0.992	0.812
\$-0.5\$	strict_filter	-6.789	0.006	0.001	-1.250	0.994	0.833
\$-0.5\$	no_filter	-0.419	0.003	0.000	-0.750	0.997	0.500
\$-0.5\$	flat_markdown	7.225	0.002	0.000	-0.375	0.998	0.000
\$-0.5\$	full_display	—	0.005	0.001	-1.375	0.995	0.917
\$0.0\$	strict_filter	-2.722	0.004	0.001	-1.250	0.996	0.833
\$0.0\$	no_filter	-0.564	0.002	0.000	-0.750	0.998	0.500
\$0.0\$	flat_markdown	4.186	0.001	0.000	-0.188	0.999	0.000
\$0.0\$	full_display	—	0.002	0.000	-0.875	0.998	0.583
\$+0.5\$	strict_filter	-4.935	0.003	0.000	-1.000	0.997	0.667
\$+0.5\$	no_filter	0.010	0.001	0.000	-0.625	0.999	0.417
\$+0.5\$	flat_markdown	2.340	0.000	0.000	-0.125	1.000	0.000
\$+0.5\$	full_display	—	0.001	0.000	-0.500	0.999	0.333
\$+1.0\$	strict_filter	-3.446	0.002	0.000	-0.500	0.998	0.333
\$+1.0\$	no_filter	0.595	0.001	0.000	-0.375	0.999	0.250
\$+1.0\$	flat_markdown	1.675	0.000	0.000	-0.062	1.000	0.000
\$+1.0\$	full_display	—	0.001	0.000	-0.375	0.999	0.250

Table 9: Policy comparison on the RC benchmark under the outside-option evaluation setup, where passengers may reject all offered bundles. Gap and win rate are computed against exhaustive full display across six split-level evaluation pairs.

Policy	Mean net profit	Gap vs full	Win rate vs full	Mean menu size	Avg meeting-point count
full display	-4377.492	–	–	2.151	1.151
strict-filter	-4373.527	3.965	0.389	1.573	0.573
heuristic					
no-filter heuristic	-4370.880	6.612	0.444	2.361	1.361

Table 10: Operational baseline comparison on the RC low-uptake benchmark. Insertion-cost greedy, minimum-latency ranking, and random-top-k provide dispatch-floor and greedy-policy context for the menu-optimization heuristic.

Policy	Mean net profit	Gap vs full	Win rate vs full	Acceptance	Menu size	Meeting-point count
full display	-4348.507	–	–	0.004	2.160	1.160
strict-filter	-4349.601	-1.094	0.306	0.005	1.577	0.577
heuristic						
insertion-cost	-4353.023	-4.516	0.306	0.004	2.017	1.017
greedy						
minimum-	-4354.325	-5.818	0.278	0.004	2.017	1.017
latency ranking						
random-top-k	-4352.281	-3.774	0.333	0.004	2.017	1.017
floor						

Table 11: Austin (Amazon Last Mile) Main Policy Comparison

Policy	Mean net profit	Gap vs full	Win rate vs full	Mean menu size	Avg meeting-point count
full display	-2992.645	–	–	9.037	8.037
strict-filter	-2998.841	-6.196	0.333	2.458	1.458
heuristic					
no-filter heuristic	-2960.523	32.122	0.833	2.764	1.764

Table 12: Seattle (Amazon Last Mile) Main Policy Comparison

Policy	Mean net profit	Gap vs full	Win rate vs full	Mean menu size	Avg meeting-point count
full display	-3149.981	–	–	5.354	4.354
strict-filter	-3103.059	46.921	1.000	1.658	0.658
heuristic					
no-filter heuristic	-3098.024	51.957	0.917	2.315	1.315

Table 13: City-level management summary for the no-filter heuristic against full display and the strict-filter heuristic.

City	No-filter vs full gap	No-filter vs strict gap	Acceptance delta	Consumer-surplus delta
Austin	32.122	38.318	-0.118	0.224
Seattle	51.957	5.035	-0.047	0.132
pooled	42.039	21.677	-0.083	0.178

Table 14: Split-level paired gap summary for the strict-filter and no-filter heuristics against full display. The inferential unit is the split-level paired mean net-profit gap; RC uses six split pairs, while Austin and Seattle each use two split pairs and are therefore descriptive only. Austin and Seattle each have only two split pairs; their gaps are reported as descriptive checks, not stable interval estimates.

Study	Policy	Mean gap	Split-level range	Range type	Positive-split share	Gap as % of [full]	Sampling unit
RC	outside- strict-filter heuristic	3.97	[-6.46, 12.07]	Split-level range (6 pairs)	0.83	0.09	6 split pairs
option							
RC	outside- no-filter heuristic	6.61	[1.41, 11.01]	Split-level range (6 pairs)	0.83	0.15	6 split pairs
option							
Austin	strict-filter heuristic	-6.20	[-26.25, 13.86]	Two-pair descriptive range	0.50	-0.21	2 split pairs (descriptive)
Austin	no-filter heuristic	32.12	[24.05, 40.20]	Two-pair descriptive range	1.00	1.07	2 split pairs (descriptive)
Seattle	strict-filter heuristic	46.92	[43.06, 50.78]	Two-pair descriptive range	1.00	1.49	2 split pairs (descriptive)
Seattle	no-filter heuristic	51.96	[50.26, 53.65]	Two-pair descriptive range	1.00	1.65	2 split pairs (descriptive)

Table 15: Benchmark-role bridge summary separating the degenerate RC mechanism benchmark from the city impact benchmarks. Gate status is evaluated on the full-display reference and the no-filter heuristic; reference-pass-only means the benchmark itself is behaviorally non-degenerate even if the no-filter policy falls outside the gate.

Study	Role	Gate status	Full acceptance	No-filter acceptance	Full opt-out	No-filter gap	Gap as % of [full]	Bridge use
RC	outside- mechanism	degenerate	0.001	0.000	0.994	6.61	0.15	mechanism only
option								
Austin	impact reference	pass only	0.161	0.044	0.694	32.12	1.07	primary impact
Seattle	impact reference	pass only	0.060	0.013	0.850	51.96	1.65	supporting check

Table A.16: MNL-regime sensitivity summary across low, medium, and higher-uptake RC regimes. Low-uptake rows diagnose mechanism behavior under near-degenerate demand; medium and higher rows stress-test the same menu and pricing rules under progressively richer behavioral variation.

Regime	Policy/pricing rule	Evidence status	Gap vs full	Acceptance	Non-home acc.	Surplus	Floor hit
low	strict-filter heuristic	current evidence	-4.591	0.003	0.001	-1.149	0.778
low	no-filter heuristic	current evidence	-4.262	0.003	0.000	-0.938	0.625
low	flat markdown pricing	current evidence	0.645	0.001	0.000	-0.375	0.000
medium	strict-filter heuristic	current evidence	14.717	0.029	0.007	-1.500	1.000
medium	no-filter heuristic	current evidence	25.969	0.027	0.000	-1.422	0.966
medium	flat markdown pricing	current evidence	44.695	0.024	0.006	-0.750	0.000
higher	strict-filter heuristic	current evidence	-17.822	0.147	0.032	-1.378	0.940
higher	no-filter heuristic	current evidence	-0.999	0.135	0.017	-1.394	0.941
higher	flat markdown pricing	current evidence	133.270	0.126	0.031	-0.750	0.000

Table A.17: Filter-validity diagnostics on the RC mechanism benchmark. Bias is mean signed error (predicted – actual); positive values indicate over-prediction. P50/P90/P95 are quantiles of displayed-offer absolute errors (seconds). FN breakdown shows false-negative pruning by meeting-point distance band: near (<500 m), mid (500–1500 m), far (>1500 m).

Signal	FN rate	ETA Bias	ETA MAE	ETA P50	ETA P90	ETA P95	IVT Bias	IVT MAE	IVT P50	IVT P90	IVT P95	FN Near	FN Mid	FN Far	Gap vs full
deployed blended ETA	0.884	2953	2953	3069	3619	3631	-2560	2560	2519	3373	3373	0.011	0.130	8.698	-12.15
heuristic ETA	0.857	0	0	0	0	0	-2104	2104	2255	3344	3373	0.035	0.118	8.421	5.01
stronger calibrated ETA	0.860	500	500	545	722	727	-2160	2160	2286	3373	3373	0.035	0.113	8.449	-25.45
oracle ETA filter proxy	0.857	2448	2448	2621	3595	3612	-2047	2047	2243	3305	3373	0.035	0.118	8.421	6.06

Table C.18: Archival single-split sensitivity diagnostic from an earlier pipeline configuration. Baseline gap in that setting is +659.6; the table is included only to show relative directional sensitivity within that historical setup, not to support the main claims of the paper.

Parameter	Description	–50%	Baseline	+50%
μ_m	Margin weight	659.6	657.7	663.2
μ_w	Walking-distance weight	663.2	657.9	661.5
μ_t	Pickup-time deviation weight	656.0	657.7	659.4
μ_v	In-vehicle time weight	659.6	659.6	659.6
μ_r	Route-delay penalty weight	578.0	659.6	657.5
μ_q	Capacity-risk penalty weight	597.5	659.6	645.6
k	Menu size cap ($k = 2, 3, 5$)	607.8	659.6	637.8

Table D.19: Predictor MAE on held-out buffer data (mean $\pm$ std across 6 RC splits). CNN-2d reduces cost MAE by 15.3% relative to the naive mean baseline. ETA and IVT CNN MAE is higher than the naive baseline because the CNN is trained on noisy online data; in deployment, ETA and IVT predictions are blended with heuristic estimates (blend weight 0.35 CNN, 0.65 heuristic).

Predictor	Cost MAE (monetary units)	ETA MAE (seconds)	IVT MAE (seconds)
CNN-2d	5.00 ± 0.59	$5,703 \pm 286$	$5,275 \pm 247$
Naive mean	5.91 ± 0.57	$1,863 \pm 66$	$1,860 \pm 67$

Table E.20: RC outside-option split-level paired gaps against full display. Within-split CI half-width is computed from episode-level pairing inside each split; main-text uncertainty uses the split-level paired means rather than pooled episode rows.

Split pair	Strict gap	Strict within-split CI/2	No-filter gap	No-filter within-split CI/2
0->1	0.35	19.60	5.08	10.97
0->1 (seed 2025)	6.78	21.15	7.63	24.29
1->0	-19.85	13.06	-5.31	10.82
2->3	16.06	20.43	13.19	19.96
3->2	13.32	17.64	6.12	14.08
4->2	7.13	30.09	12.96	23.57

Table E.21: Austin and Seattle split-level paired gaps against full display. Each city has only two split pairs, so these rows should be read descriptively rather than as stable interval estimates.

City	Split pair	Strict gap	No-filter gap
Austin	0->1	-26.25	24.05
Austin	1->0	13.86	40.20
Seattle	0->1	50.78	53.65
Seattle	1->0	43.06	50.26

Table E.22: City-level policy-conditioned displayed-offer diagnostics. Displayed ETA/IVT MAE and false-negative pruning are diagnostics on displayed offers rather than direct welfare outcomes; acceptance rate and consumer surplus are realized passenger-facing outcomes.

City	Policy	FN pruning	Displayed ETA MAE	Displayed IVT MAE	Acceptance rate	Consumer surplus
Austin	full display	0.598	598.5	726.8	0.161	-1.227
Austin	strict-filter heuristic	0.598	570.8	707.2	0.043	-1.102
Austin	no-filter heuristic	0.000	802.5	951.0	0.044	-1.003
Seattle	full display	0.782	943.3	727.2	0.060	-1.315
Seattle	strict-filter heuristic	0.782	908.4	689.8	0.010	-1.194
Seattle	no-filter heuristic	0.000	3847.2	2502.7	0.013	-1.183

Table F.23: Computational cost summary for the RC instance (90 customers, 10 meeting points, 15 vehicles). Training and inference times are measured on a standard CPU (no GPU used). HGS time limits are upper bounds configured per episode; actual routing time is typically below the limit.

Phase	Metric	Value
Predictor training	Training episodes	12
	Steps per episode	≈ 700
	Wall-clock per episode	≈ 26.5 s
	Total training time	≈ 318 s (≈ 5.3 min)
	Initial phase epochs	12
	Initial phase total time	≈ 0.4 s
Menu-construction inference	Mean time per request	44.9 ms
	(CNN forward pass + insertion cost)	
HGS routing	Reopt time limit (per step)	1.1 s
	Final reopt time limit	1.5 s
	Mean step time (menu + HGS)	≈ 43 ms